

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40017
Subject Title	Experiment X
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	AP30022
Objectives	The course introduces various advanced technologies used for the next generation artificial intelligence (AI) including machine learning (ML) and robotics involving computer vision, and computer simulation. Students can also understand the concepts of engineering design process by implementing a product design.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) apply the general operational principles of forward kinematics and inverse kinematics to a robotic arm; b) utilize and understand the general operational principles of machine learning to practical experiments; c) apply and understand computer vision to object detection; d) utilize the concepts of computational physics and dynamical simulation to off-line programming and digital twins; e) apply the knowledge to solve some real-life problems.
Subject Synopsis/ Indicative Syllabus	Robotics: Forward kinematics, inverse kinematics, industrial robot, robotic arm, robot calibration, robotics simulator, off-line programming Artificial intelligence (AI): Applications of artificial intelligence, philosophy of artificial intelligence, history of artificial intelligence Machine learning (ML): Timeline of machine learning, data mining, supervised learning, statistical classification, regression analysis, cluster analysis, dimensionality reduction, structured prediction, anomaly detection, neural network, reinforcement learning Computer vision: Image sensor, digital image processing, image analysis, object detection, activity recognition, video tracking Scientific modelling (Simulation): Mathematical model, computational physics, computer experiment, computer simulation, dynamical simulation, verification and validation of computer simulation models, performance tuning, uncertainty quantification, safety engineering, human-in-the-loop, visual effects, digital twin Engineering design process: Decision-making, conceptual design, reverse engineering, requirements analysis

Teaching/Learning Methodology	Lecture: Background knowledge behind all experiments will be systematically introduced in lectures. Class work and assignments related to the content of lectures will be used to enhance students learning. Student-centred tutorial: Tutorial classes allow students to ask questions related to abstract and difficult learning content, followed by in-depth explanation and discussion, resulting in fostering their skills in analytical power, concept acquisition, critical thinking, problem-solving and life-long learning. Lab session: Experiments are essential for students to relate the concepts to practical applications and they are exposed to hands-on experience and proper use of equipment and also analytical skills on interpreting experimental results.																																								
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Continuous assessment</td><td>50</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Practical test</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>3. Written test</td><td>20</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100</td><td colspan="5"></td></tr></table> Continuous assessment: Continuous assessment includes assignments. Assignments will strengthen the students’ basic knowledge and the analytical skills to solve the problems related to this subject. Practical test: Practical test will review students’ understanding of the course and assess their ability to solve problems, in particular using robot software and computer simulation Written test Written test is useful to assess students’ knowledge learned from the lectures and laboratory works.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Continuous assessment	50	✓	✓	✓	✓	✓	2. Practical test	30	✓	✓	✓	✓		3. Written test	20	✓	✓	✓	✓	✓	Total	100					
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2. Practical test	30	✓	✓	✓	✓																																				
3. Written test	20	✓	✓	✓	✓	✓																																			
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Student Study Effort Expected	<table><tr><td>Class contact:</td><td></td></tr><tr><td>• Lecture & tutorial</td><td>13 Hrs.</td></tr><tr><td>• Lab</td><td>39 Hrs.</td></tr><tr><td>Other student study effort:</td><td></td></tr><tr><td>• Self-study</td><td>68 Hrs.</td></tr><tr><td>Total student study effort</td><td>120 Hrs.</td></tr></table>	Class contact:		• Lecture & tutorial	13 Hrs.	• Lab	39 Hrs.	Other student study effort:		• Self-study	68 Hrs.	Total student study effort	120 Hrs.																												
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Reading List and References	• Simon Rogers & Mark Girolami, <i>A First Course in Machine Learning</i> • https://aeturrell.github.io/python4DS/welcome.html • https://www.youtube.com/watch?v=3Gs4u4ixsyw																																								